1 A kernel over subsets

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1.

$$k(A,B) = \begin{cases} 0 & \text{if } A \cup B = \emptyset \\ \dfrac{|A \cap B|}{|A \cup B|} & \text{otherwise} \end{cases} = \begin{cases} 0 & \text{if } B \cup A = \emptyset \\ \dfrac{|B \cap A|}{|B \cup A|} & \text{otherwise} \end{cases} = k(B,A)$$

Hence K is symmetric.

2. $A_1 = \emptyset$, $A_2 = \{x_1\}$, $A_3 = \{x_2\}$,$A_4 = \{x_1, x_2\}$

$$K = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} & 1 \end{pmatrix}$$

3. Yes, for $n = 2$ k can be expressed as $k(A_i, A_j) = \langle \phi(A_i), \phi(A_j) \rangle$.

By the M.A. theorem this kernel is p.d. and has a corresponding RKHS. The correct mappinps, $\phi(A_i)$ are printed in the jupyter code below:

K_decompostion

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[ ]:
```python
import numpy as np
np.set_printoptions(precision=1)
def phi(i):
    M=np.array([[1,0,.5],[0,1,.5],[.5,.5,1]])
    eigenvalues,eigenvectors=np.linalg.eig(M)

    return np.array([0, np.sqrt(eigenvalues[0])*eigenvectors[i,0],np.
 ↪sqrt(eigenvalues[1])\
        *eigenvectors[i,1],np.sqrt(eigenvalues[2])*eigenvectors[i,2]])
Phi=[np.zeros([1,4]),*[phi(i) for i in range(3)]]
K=np.zeros([4,4])
for i in range(4):
    for j in range(4):
        K[i,j]=np.round(np.dot(np.array(Phi[i]),np.array(Phi[j]).T),4)
```

[ ]:
```python
print(Phi)
```

```
[array([[0., 0., 0., 0.]]), array([ 0. , -0.7, -0.7, -0.3]), array([ 0. , -0.7,
0.7, -0.3]), array([ 0.0e+00, -9.2e-01, -4.1e-16,  3.8e-01])]
```

[ ]:
```python
print(K)
```

```
[[ 0.   0.   0.   0. ]
 [ 0.   1.  -0.   0.5]
 [ 0.  -0.   1.   0.5]
 [ 0.   0.5  0.5  1. ]]
```